

ME-502: ANM Assignment 3



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2025MEM1003

The assignment questions-

You are required to write a program to solve the following second order differential equation.

$$mx'' + cx' + kx = f(t)$$

Answers-

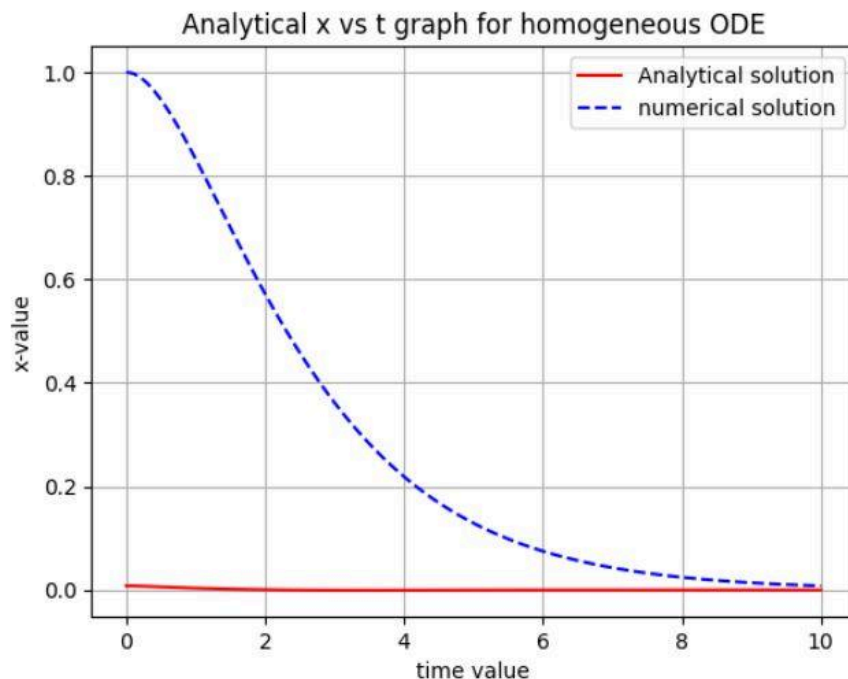
For testing purpose, we shall keep the input values same as given below-

```
Welcome to 2nd order ODE solver [mx''+cx'+kx=f(t)], by using generalized alpha method, please specify your values!  
Enter your initial value x0: 1  
Enter your initial differential value diff_x0: 0  
Enter your mass: 10  
Enter your dampening constant: 15  
Enter your spring constant: 12  
Enter your learning rate/alpha: 0.5  
Enter your time step: .1  
Enter your initial time: 0  
Enter your final time: 10
```

For homogeneous case-

6- For validation of a homogeneous case, you can compare the solution obtained by your program with the analytical one. You are required to validate your program for a homogeneous case with non-zero initial conditions. The analytic solution for this case has been provided at the end of assignment.

Output-



Formulation of Generalized- α method for 2nd ODE-

Given, $m\ddot{x} + (\dot{x} + kx) = f(t)$

where, $\dot{x}_0 = \text{known}$
 $x_0 = \text{known}$

Let $\dot{x} = y_2$ to $y_1 = y_2$
 $x = y_1$

Remember,

$\dot{x}_0 = y_2^{(0)} = \text{known}$
 $x_0 = y_1^{(0)} = \text{known}$

Further notations,

$t_0 = \text{Initial time}$
 $t_f = \text{Final time}$
 $\Delta t = \text{time step}$

Now, Generalized α -algorithm $\rightarrow$

$$\frac{y_{n+1} - y_n}{\Delta t} = \alpha f(y_{n+1}, t_{n+1}) + (1-\alpha) f(y_n, t_n)$$

we have,

$$m \dot{y}_2 + c y_2 + k y_1 = f(t)$$

$$\rightarrow \dot{y}_2 = \frac{f(t) - (c y_2 + k y_1)}{m} \quad - (1)$$

$$y_1 = y_2 \quad - (2)$$

P.T.O

$$\begin{array}{l|l} y_2^{(0)} = k & y_2^{(1)} = 0 + \\ y_1^{(0)} = k & y_2^{(1)} = 0 + k \end{array}$$

classmate

Date _____
Page _____

For ①

$$\frac{y_2^{(1)} - y_2^{(0)}}{\Delta t} = \alpha \left[\frac{f(t^{(0)}) - (y_2^{(1)} - k y_1^{(0)})}{m} \right]$$

$$+ (1-\alpha) \left[\frac{f(t^{(0)}) - (y_2^{(0)} - k y_1^{(0)})}{m} \right]$$

③

$$\left. \begin{array}{l} y_2^{(1)} \\ y_1^{(1)} \end{array} \right\} \rightarrow \text{unknown}$$

For ②

$$\frac{y_1^{(1)} - y_1^{(0)}}{\Delta t} = \alpha [y_2^{(1)}] + (1-\alpha) [y_2^{(0)}]$$

$$y_1^{(1)} = \alpha \Delta t [y_2^{(1)}] + (1-\alpha) \Delta t [y_2^{(0)}] + y_1^{(0)} \quad \text{--- ④}$$

Remember, $\dot{x} = y_2$

$x = y_1 \rightarrow$ we need this
or $x^{(n+1)} = y_1^{(n+1)}$

But eqn ④ has $y_2^{(1)}$ unknown term.

so using ③ [also $t^{(1)} = t^{(0)} + \Delta t$]

$$y_2^{(1)} = \frac{\alpha \Delta t}{m} [f(t^{(0)} + \Delta t) - (y_2^{(1)} - k y_1^{(1)})]$$

$$+ \frac{(1-\alpha) \Delta t}{m} [f(t^{(0)}) - (y_2^{(0)} - k y_1^{(0)})] + y_2^{(0)}$$

Further,

$$y_2^{(1)} \left(1 + \frac{\alpha \Delta t}{m} \right)$$

$$= \frac{\alpha \Delta t}{m} (f(t^{(0)} + \Delta t) - k y_1^{(1)})$$

$$\left(\frac{1 - \alpha \Delta t}{m} [f(t^{(0)}) - (y_2^{(0)} - k y_1^{(0)})] + y_2^{(0)} \right)$$

$$\checkmark y_2^{(1)} = \left[\frac{\alpha \Delta t}{m + \alpha \Delta t} [f(t^{(0)} + \Delta t) - k y_1^{(1)}] \right.$$

$$+ \left[\frac{(1 - \alpha) \Delta t}{m + \alpha \Delta t} [f(t^{(0)}) - (y_2^{(0)} - k y_1^{(0)})] \right]$$

$$+ \frac{y_2^{(0)} (m)}{(m + \alpha \Delta t)} \quad \text{--- (5)}$$

Sub (5) in (4)

$$y_1^{(1)} = \alpha \Delta t \left\{ \left[\frac{\alpha \Delta t}{m + \alpha \Delta t} [f(t^{(0)} + \Delta t) - k y_1^{(1)}] \right. \right.$$

+ --- + ---

collect $y_1^{(1)}$ on one side.

$$y_1^{(1)} + k \frac{(\alpha \Delta t)^2}{m + \alpha \Delta t} y_1^{(1)} = \text{---}$$

or
rather solve it is $y_2^{(1)}$

sub ④ in ⑤

$$y_2^{(2)} = \left[\frac{\alpha \Delta t}{m + c \Delta t} \right] \left[f(t^{(1)} + \Delta t) - k \right]$$

$$\left(\alpha \Delta t + [y_2^{(1)}] + (1 - \alpha) \Delta t [y_2^{(0)} + y_1^{(0)}] \right)$$

$$+ \left[\frac{(1 - \alpha) \Delta t}{m + c \Delta t} \right] \left[f(t^{(0)}) - (y_2^{(0)} - k y_1^{(0)}) \right]$$

$$+ \left[\frac{y_2^{(0)} m}{m + c \Delta t} \right]$$

$$y_2^{(1)} + k \frac{(\alpha \Delta t)^2}{m + c \Delta t} (y_2^{(1)}) \approx \sim$$

$$= y_2^{(1)} \left(1 + \frac{k (\alpha \Delta t)^2}{m + c \Delta t} \right)$$

$$\therefore y_2^{(1)} = \frac{1}{\left[1 + \frac{k (\alpha \Delta t)^2}{m + c \Delta t} \right]} \left[\left[\frac{\alpha \Delta t}{m + c \Delta t} \right] \left[f(t^{(0)} + \Delta t) - k \right] \right.$$

$$\left. - k (1 - \alpha) \Delta t [y_2^{(0)} + y_1^{(0)}] \right]$$

$$+ \left[\frac{(1 - \alpha) \Delta t}{m + c \Delta t} \right] \left[f(t^{(0)}) - (y_2^{(0)} - k y_1^{(0)}) \right]$$

$$+ \left[\frac{y_2^{(0)} m}{m + c \Delta t} \right]$$

— ⑥

Observation-

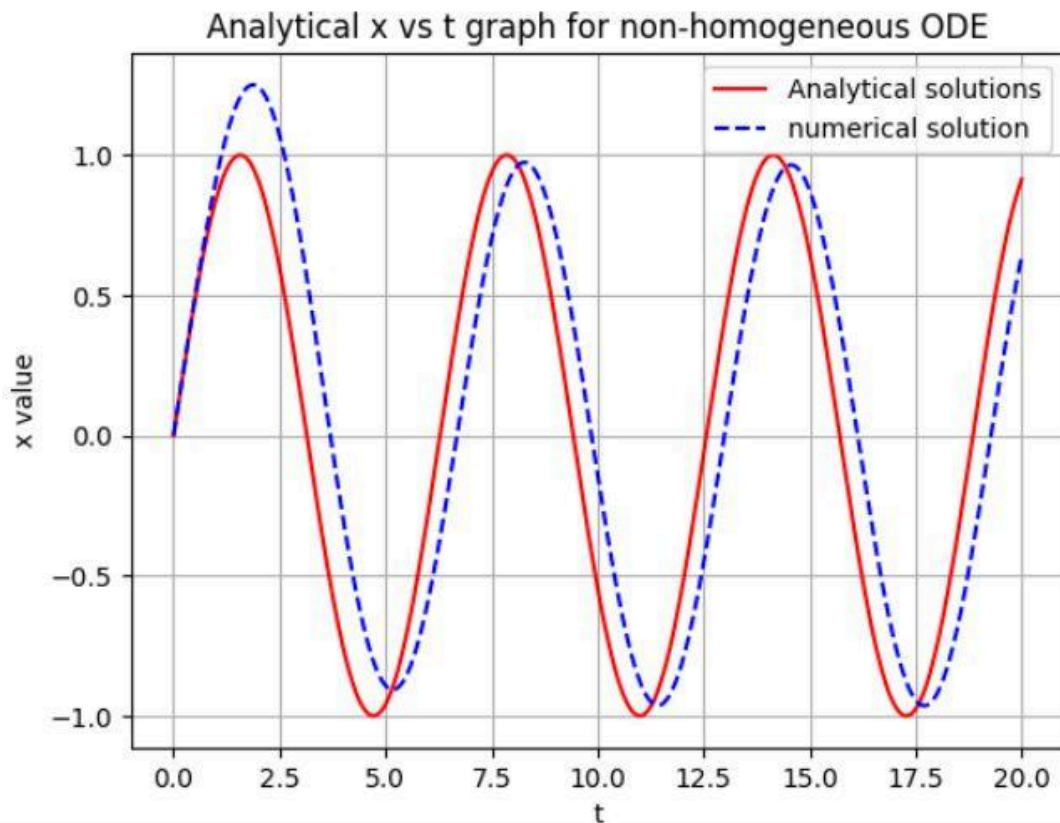
1. The analytical solution is given by $x(t) = e^{-\psi\omega t}(A\sin(\omega_d t) + B\cos(\omega_d t))$ with the same initial conditions as numerical for solving.
2. Both curves show a decaying oscillatory behavior, as expected for a damped homogeneous system.
3. Solution successfully converges after certain time intervals. Though, major differences appear due to numerical time-stepping errors or finite step size effects.
4. The numerical curve remains bounded and decays smoothly without divergence, indicating that the chosen integration scheme is stable for this ODE.

7- For validation of non-homogeneous cases, we are going to try the cool method of manufactured solution. In this technique, we are going to first assume a simple solution (for example: $x = \sin(t)$), and corresponding to it we are going to find out the $f(t)$ (from differential equation) and initial conditions. Now we can use our program with the obtained force and initial conditions. The results obtained are expected to match the assumed solution.

Input-

```
Welcome to 2nd order ODE solver [mx''+cx'+kx=f(t)], by using generalized alpha method, please specify your values!  
Enter your initial value x0: 0  
Enter your initial differential value diff_x0: 1  
Enter your mass: 10  
Enter your dampening constant: 15  
Enter your spring constant: 12  
Enter your learning rate/alpha: 0.5  
Enter your time step: 0.01  
Enter your initial time: 0  
Enter your final time: 20
```

Output-



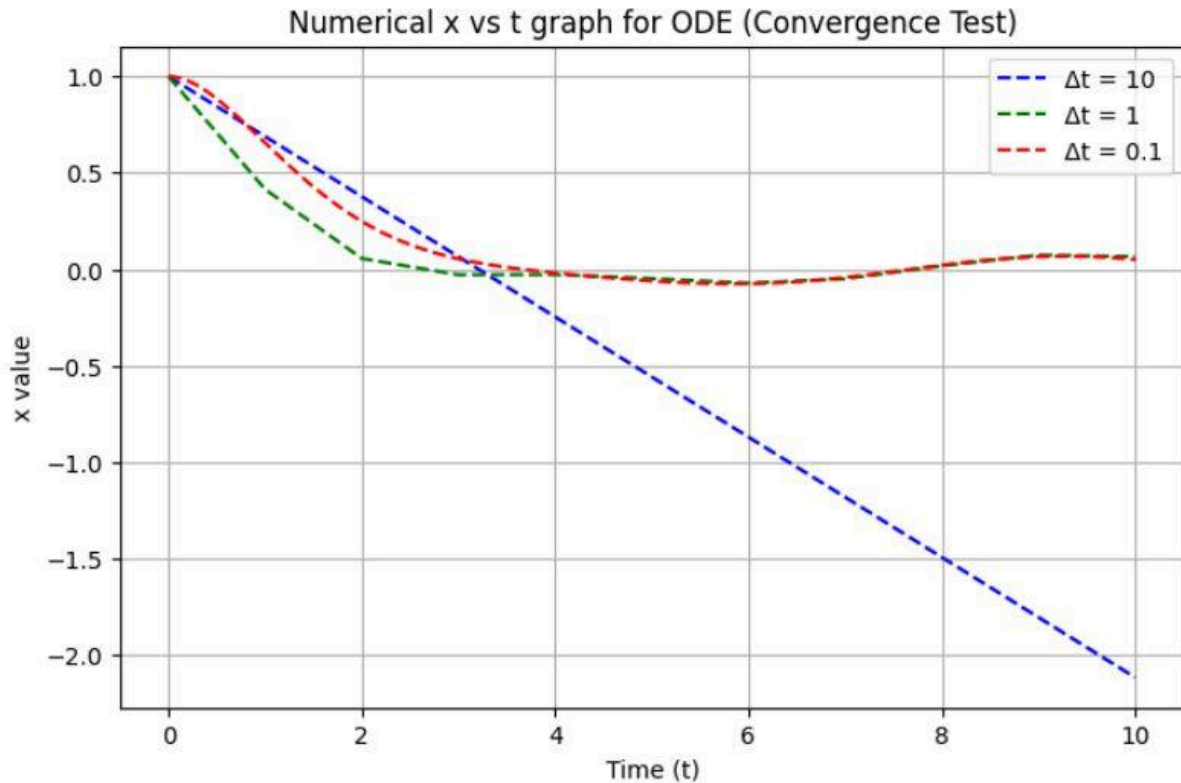
Observation-

1. On this case, a simple analytical (manufactured) solution was assumed as $x(t)=\sin(t)$ for the sake of testing our solver.
2. Both solutions exhibit oscillatory motion of similar frequency and amplitude.

3. The small phase difference observed is due to the finite time-step size used in the numerical integration. Which can be decreased for smaller time steps.

8- Solve the system for an arbitrary function and initial conditions. Hint: In order to make sure that your results are correct, run the program with successive decreasing values of Δt , and make sure that after a while your results do not change by further reduction in Δt value.

Note- Three different inputs with different timesteps taken for analysis-

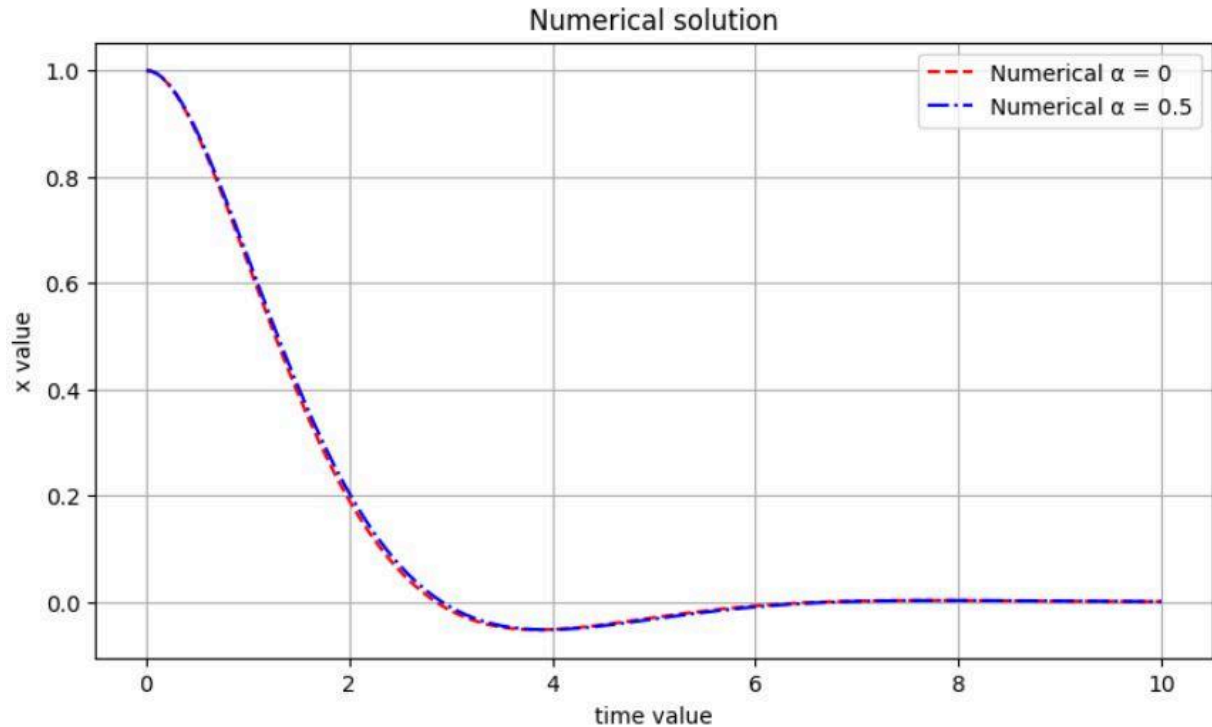


Observation-

1. For a large time step ($\Delta t=10$), the numerical solution diverges significantly and shows unrealistic behavior.
2. As the time step is reduced to $\Delta t=1$, the solution becomes more accurate but still exhibits some deviation from the expected smooth behavior.
3. With a smaller $\Delta t=0.1$, the numerical solution becomes smooth, stable, and physically realistic, indicating convergence toward the true solution.
4. The accuracy of the generalized-alpha method improves as Δt decreases. Though, after certain reduction of time steps the results do not change.
5. Large Δt values can make the scheme unstable or overdamped.
6. Smaller Δt values ensure that the algorithm remains stable within the problem's physical damping limits.

9- Try any two values of α from 0, 0.5, and 1 and compare the performance of these two methods. You can carry out the comparison in terms of the accuracy and stability.

Output-



Observations-

1. For $\alpha = 0$, the solution exhibits slightly higher amplitude in the early oscillations, meaning it is less numerically dissipative.
This case is more accurate for short times but may develop small numerical oscillations for coarser time steps.
2. For $\alpha = 0.5$, the response is slightly smoother and shows additional numerical damping, which improves stability but introduces a minor phase lag.
This makes the $\alpha = 0.5$ method more suitable for long-term simulations or when large time steps are used.